# 🗂️Bash File Commands Cheat Sheet

This is a **focused cheat sheet for file operations** in Bash, including **file tests and shell scripting commands**.

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## ✂️LIST FILES AND DIRECTORIES

```
ls                  # List files
ls -l               # Detailed listing
ls -a               # Include hidden files
ls -lh              # Human-readable sizes
ls -R               # Recursive listing
```

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## ✒️CURRENT DIRECTORY

```
pwd                 # Print working directory
cd /path/to/dir     # Change directory
cd ~                # Go to home directory
cd ..               # Move up one directory
```

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## 🛏️CREATE FILES AND DIRECTORIES

```
touch file.txt      # Create empty file
mkdir folder_name   # Create directory
mkdir -p dir1/dir2  # Create nested directories
```

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## 🌂REMOVE FILES AND DIRECTORIES

```
rm file.txt         # Remove file
rm -i file.txt      # Remove with confirmation
rm -r folder        # Remove folder recursively
rmdir folder        # Remove empty folder
```

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## 🌡️ COPY AND MOVE FILES

```
cp source.txt dest.txt       # Copy file
cp -r dir1/ dir2/            # Copy directory recursively
mv old.txt new.txt           # Rename or move file
```

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## 🎒VIEW FILE CONTENTS

```
cat file.txt           # Display file
less file.txt          # Scrollable viewer
head -n 5 file.txt     # First 5 lines
tail -n 5 file.txt     # Last 5 lines
tail -f logfile.log    # Follow file in real-time
```

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## 🎓FILE INFORMATION

```
file file.txt          # Show file type
stat file.txt          # Detailed info (size, modification date)
ls -l file.txt         # Permissions, owner, size, date
du -h file.txt         # Disk usage
```

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## 🎩FILE PERMISSIONS

```
chmod 755 file.txt     # Change permissions (rwxr-xr-x)
chmod +x script.sh     # Make executable
chown user file.txt    # Change owner
chgrp group file.txt   # Change group
```

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## 🎽SHELL SCRIPT FILE TESTS (CONDITIONALS)

```
# Check if file exists and is a regular file
if [ -f file.txt ]; then echo "File exists"; fi

# Check if directory exists
```

```
if [ -d folder ]; then echo "Directory exists"; fi

# Check if file exists (any type)
if [ -e file.txt ]; then echo "Exists"; fi

# Check read/write/execute permissions
[ -r file.txt ] && echo "Readable"
[ -w file.txt ] && echo "Writable"
[ -x file.txt ] && echo "Executable"
```

Other useful tests:

```
[ -s file.txt ]  # Non-empty file
[ -L link ]      # Symbolic link
```

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## 1 🏮 SEARCH AND FILTER FILES

```
grep "pattern" file.txt       # Search text
grep -i "pattern" file.txt    # Case-insensitive
grep -n "pattern" file.txt    # Show line numbers
grep -c "pattern" file.txt    # Count matches
find /path -name "*.txt"      # Find files by name
locate file.txt                # Quickly locate files
```

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## 1✂️ COUNT WORDS / LINES / CHARACTERS

```
wc file.txt              # lines words chars
wc -l file.txt           # lines
wc -w file.txt           # words
wc -c file.txt           # bytes
```

Examples:

```
ls -l | wc -l            # Count files in directory
grep -c "error" file.txt  # Count pattern matches
```

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## 1✒️REDIRECTION & PIPES

```
command > file.txt       # Overwrite output
command >> file.txt      # Append output
command < file.txt       # Input from file
cmd1 | cmd2              # Pipe output of cmd1 to cmd2
```

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## 1🛏️ARCHIVE AND COMPRESS FILES

```
tar -cvf archive.tar folder/    # Create archive
tar -xvf archive.tar             # Extract archive
gzip file.txt                     # Compress
gunzip file.txt.gz                # Decompress
zip archive.zip file1 file2       # Zip files
unzip archive.zip                 # Unzip
```

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## 1🌂SYMBOLIC LINKS

```
ln -s target link_name    # Create symbolic link
ls -l                     # Show links with ->
```

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## 1 🌡️ FILE COMPARISON

```
diff file1.txt file2.txt   # Show differences
cmp file1.txt file2.txt    # Compare byte by byte
```

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### 💡EXAM & LAB TIPS

- Use `-f` to check **regular files** in scripts.
- Use `-d` to check **directories**.
- Combine tests: `[ -f file.txt ] && [ -r file.txt ]` → exists & readable
- Use `ls -lh` for human-readable file sizes.
- Always quote filenames with spaces: `"$file"`